UIC Department of Chemical Engineering
UNIVERSITY OF ILLINOIS AT CHICAGO
COLLEGE OF ENGINEERING

TEAM ASSIGNMENT #5

Course: CHE 433. PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2019
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name:____

Due Date: Friday November 1st by 5:00pm in my office

Number of points............................100
Your score..................................____

Note: **All problems are worth 100 points. You final score will be the average value**

Problem 5A: Cyclohexane production

For the cyclohexane production process of **Assignment #4**, consider the option of increasing the cyclohexane recovery by replacing the low pressure separator with a distillation column. Design the distillation column to maximize the recovery of cyclohexane and update your simulation. Recalculate your fixed capital investment and the operating costs. Update the list of all equipment (except the reactor) and recalculate the total capital cost. Update the list of all utilities and calculate the total operating cost, including the cost of raw materials, revenues from the product and credits for fuel. Start again with your own simulation from **Assignment #4**. Plant capacity is 300 KMTA of cyclohexane with plant operating data defined in **Assignment #3**. Cooling water is available by a nearby cooling tower that has been designed for dry bulb temperature of 85 deg F, relative humidity of 70% and an approach temperature of 10 deg C. The maximum hot water temperature was taken as 45 deg C for the design of the cooling tower. Multiple pressure levels of steam are available by a steam plant nearby.

REQUIRED REPORT SECTIONS

- Outline your steps you followed to reach your final column design
- Final process flowsheet printout
- ASPEN report for the distillation column
- Heat and Weight Balance for the entire unit
- List of all equipment for with the capital costs (see example below)
- List of utilities, raw materials and products with operating costs

Cyclohexane Production Cost Estimate

EQUIPMENT LIST: CAPITAL COSTS

	Project:	Assignment #2	*Chemical Engineers:*			
	Location:	Chemical Engineering Department	*Date:*			
Equipment Class	**ASPEN BlockID**	**Service Description**	**Tag**	**Material Cost**	**Installed Cost**	**Annual Cost, $/yr**
Distillation						
Towers	xxxx	Finishing Column	T-1	$1,950,958	$4,095,200	$819,040
Towers	xxxx	Packing, Trays, Internals	T-1			
Drums						
Vertical Tank	xxxx	Seperation Drum	V-1			
Vertical Tank	xxxx	Reflux Drum	V-2			
Heat Exchangers						
S&T Exchanger		Feed-Product Exchanger	E-1			
S&T Exchanger		Product Cooler	E-2			
Air Coolers						
Air Cooler		Reflux Condenser	EA-1			
Air Cooler			EA-2			
Pumps						
Pump		Reflux Pump	P-1			
Pump			P-2			
Turbines						
Compressors						
Reactors						